

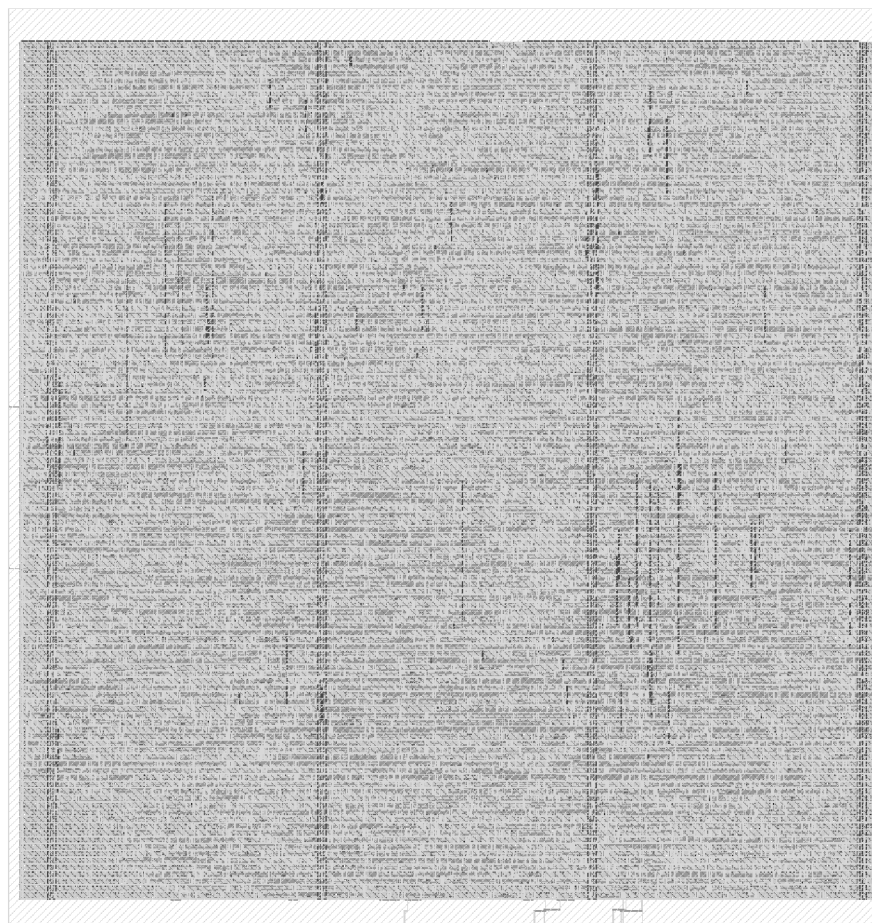


Lambda KV-Cache Coprocessor

KV-cache compression, scoring & eviction —
in silicon.

SSCS Chipathon 2026 · Track D · GF180MCU · Team A71

Chaithu Talasila



lambda_kv_coproc · GF180MCU · 0.262 mm²

The KV-cache is the fastest-growing cost in LLM decode.

Every generated token re-reads the whole key/value cache. It scales with context length and comes to dominate on-device memory and bandwidth — so the host spends its energy moving the cache, not computing.

Do the cache work at the edge of memory

Three hardened engines behind a serial loader — the host streams tokens in and reads compressed records back.

KVE · COMPRESS

CQ-3-rot value path

Walsh-Hadamard rotation + per-token amax + INT3 quantization → rotated INT3 codes and one fp16 scale per token. Synthesizable, bit-exact vs. reference.

TIU · SCORE

H2O token importance

Heavy-hitter scoring across cache slots → a per-slot keep-tier bitmap and one eviction-victim slot, so cold tokens leave first.

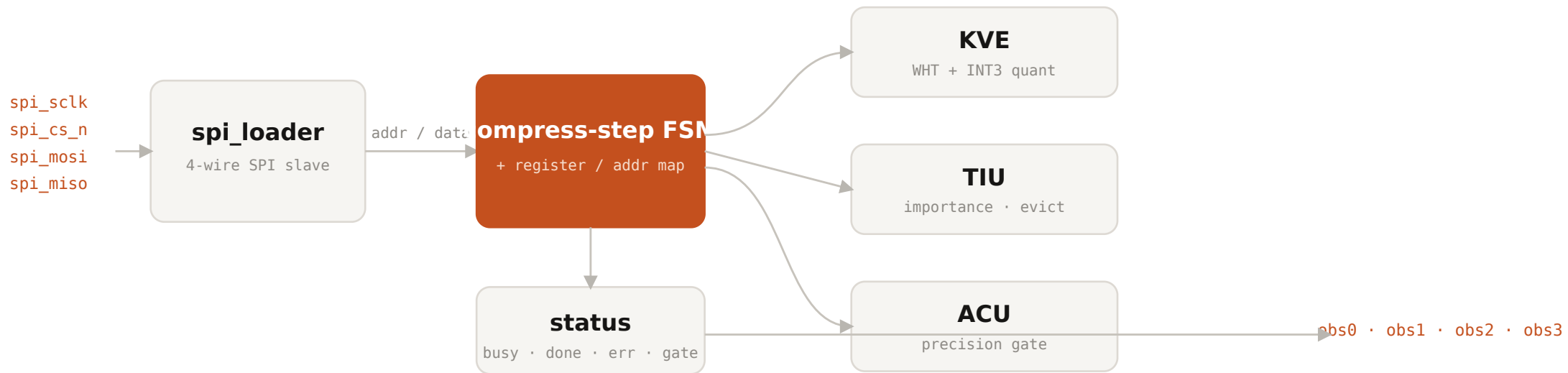
ACU · GATE

Precision controller

A divide-free per-tile gate over the importance distribution → a keep-fp16 decision when a tile is peaky enough to warrant it.

Everything streams over a 4-wire SPI slave — 6 functional pins + 4 status bits.

One serial link in, compressed records out



Host writes $V + W$, pulses START, polls STATUS, then reads CODES / SCALE / $\hat{V}$ + the decision byte – all over the same four wires.

One compress step

01

Host streams L value tokens (V) + attention masses (W)
over SPI, byte-addressed

02

KVE compresses each token
INT3 codes · fp16 scale · reconstruction $\hat{V}$

03

TIU scores importance
keep-tier bitmap + eviction victim

04

ACU gates precision
keep-fp16 decision

05

Host polls STATUS, reads back records
codes / scale / $\hat{V}$ + decision byte

BYTE-ADDRESSED REGISTER MAP

ADDR	REGION	DIR
0x0300	W — attention mass	write
0x0400	V — value tokens	write
0x0800	CODES — INT3	read
0x0A00	SCALE — fp16	read
0x0C00	$\hat{V}$ — reconstruction	read
0x0002	DECISION byte	read
0x0001	STATUS mirror	read

From **RTL** to **GDS**, driven end-to- end by an agent.

Hardened the core — LibreLane / GF180 to clean signoff.

run

Fixed the pin audit — bus obs_out[3:0] → scalar obs0..obs3; vdd/vss → VDD/VSS.

b4722ee

Reordered for the BV slot — ground moved to pin 1.

a432f94

Verified & integrated — re-hardened, checked GDS labels, reconciled the delivered padframe DEF.

21819a4

| Every step reproduces in CI — auditable, not asserted.

Clean signoff — every check at zero.

Magic DRC 0

KLayout DRC 0

Routing DRC 0

LVS 0

Antenna 0

XOR 0

Setup / hold 0

0.262_{mm²}

Die area

7,080

Standard cells

62%

Utilization

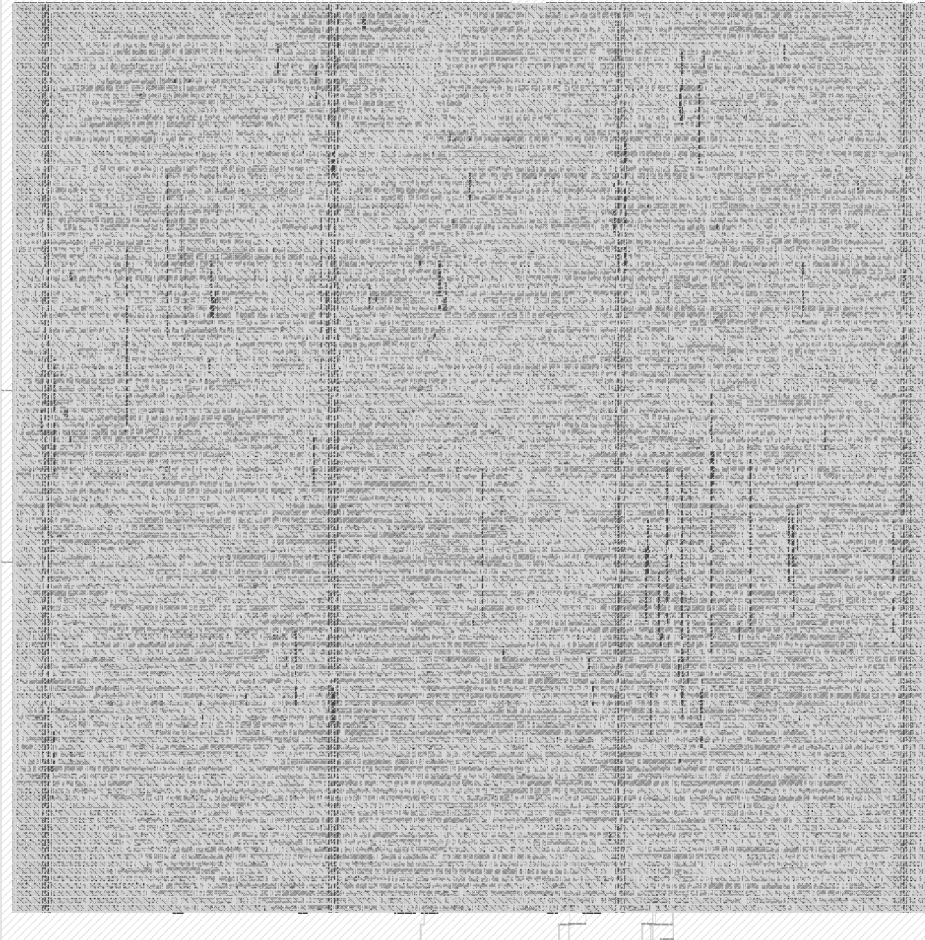
300_{ns}

Clock period

gf180mcuD · gf180mcu_fd_sc_mcu7t5v0 · LibreLane Classic, reproduced in CI.

The hardened block

07 — PHYSICAL



final_core/gds/lambda_kv_coproc.gds

- **502.89 × 520.81 μm** core, no padding — a pre-integration macro.
- **GF180MCU · 180 nm, 5 V** — mcu7t5v0 standard cells.
- **Text-labelled pins** — obs0..obs3, VDD/VSS and the SPI signals, matched by name.
- **Timing closed** at 300 ns across all signoff corners.

Slots into the org padframe by name

- We submit a **core-only** block + an `info.yaml` pin list.
- The org generates the padframe, sized to our area, and returns a **DEF**.
- Placed in a **BV slot** — 1/8, 16 pins, ground on pin 1. We use 12; unused slots become analog pads.
- Delivered **BV + BH** variants — both reconcile against our submission exactly.

BV PAD MAP — OUR 12 PINS

PIN	NAME	SLOT · TYPE
1	VSS	W12 · ground
2	clk	W13 · in_s
3–6	rst_n · spi_sclk/cs_n/mosi	W14–17
7	spi_miso	W18 · bi_t
8–11	obs0 · obs1 · obs2 · obs3	W19–22 · bi_t
12	VDD	N01 · power

Proven, and reproducible

- **Self-checking SPI test** — write W + V, START, poll STATUS, read back decision / keep-bitmap / codes / scale. All checks pass.
- **CI rebuilds it** — simulation + hardening on every push.
- **LVS** ties the layout back to the netlist — the GDS is what the RTL says.

Calibrated: verification is at the **RTL + physical-signoff** level. Gate-level simulation and exhaustive coverage are named future work, not implied here.

```
make sim-coproc · make coproc
```

Ready for the **dry-run**.

- **DRC-clean GDS ready** for the Aug 28 dry-run to the channel partner.
- **Pin map locked** against the delivered DEF — naming and order confirmed.
- **Next** — fabrication on the shared reticle, then bring-up over the SPI link.

 **LONGHORN SILICON**

Chips, designed at Texas.

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